

Cross-cultural recognition memory: intergroup correlation results

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Overview

Intergroup itemwise correlations and paired-bootstrap comparisons for two conditions (*Globalized-Music*, *Industrial-Nature*) and two trial types (hits and false alarms). All numbers were produced by the Python pipeline in `Analysis-Scripts-v2/python/`; see `STATS.md` / `stats.pdf` for the methods. Bootstrap settings: $n_{\text{boot}} = 2000$ for pairwise CIs, $n_{\text{boot}} = 5000$ for the paired test, attenuation correction in `fixed` mode, participant-level resampling, `minResp` = 2. Reported CIs are **BCa** (bias-corrected and accelerated) unless otherwise noted; percentile and Fisher- z alternatives appear in the diagnostic tables. The default p -value is the recentered-null variant.

Site partition and within-group reliabilities

Condition	Group	n_{subj}	n_{items}	$\hat{\rho}_{\text{hit}}^{\text{SB}}$	$\hat{\rho}_{\text{fa}}^{\text{SB}}$
Globalized-Music	US	96	80	0.776	0.865
Globalized-Music	SanBorja	63	80	0.716	0.741
Globalized-Music	Tsimane	68	80	0.217	0.695
Industrial-Nature	US	141	80	0.782	0.910
Industrial-Nature	SanBorja	56	80	0.707	0.802
Industrial-Nature	Tsimane	81	80	0.707	0.804

Globalized-Music

Hits

Pairwise intergroup correlations (BCa 95% CIs)

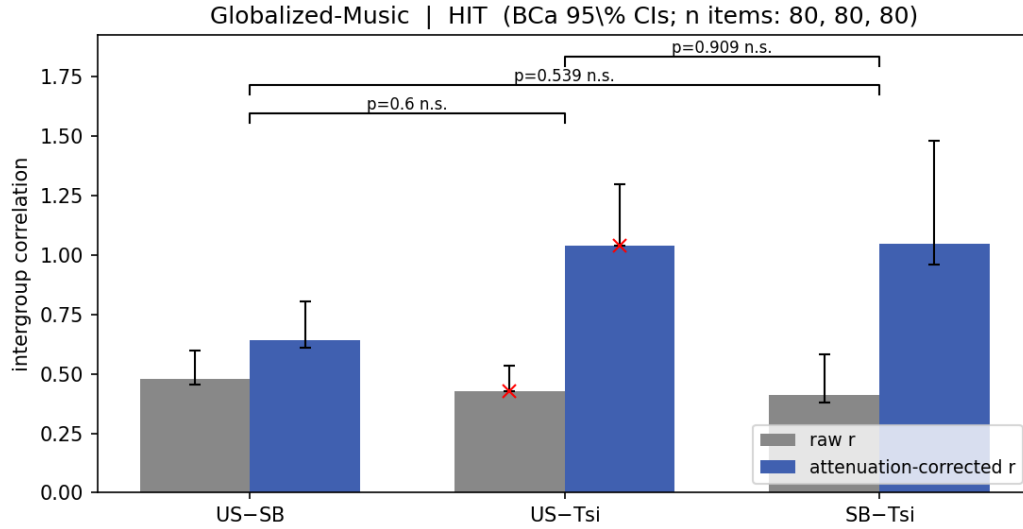
Pair	$\hat{r}$ (raw)	95% CI (BCa)	$\hat{r}^*$ (corrected)	95% CI (BCa)
US–SB	+0.478	[+0.454, +0.600]	+0.641	[+0.609, +0.805]
US–Tsi	+0.426	[+0.426, +0.533]	+1.038	[+1.039, +1.299]
SB–Tsi	+0.413	[+0.379, +0.583]	+1.047	[+0.961, +1.480]

Diagnostic — alternative CIs on corrected $\hat{r}^*$

Pair	$\hat{r}^*$	Percentile 95% CI	Fisher- z 95% CI	BCa 95% CI
US–SB	+0.641	[+0.317, +0.669]	[+0.475, +0.763]	[+0.609, +0.805]
US–Tsi	+1.038	[+0.306, +1.035]	[+1.000, +1.000]	[+1.039, +1.299]
SB–Tsi	+1.047	[+0.230, +1.120]	[+0.998, +1.000]	[+0.961, +1.480]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US–SB vs US–Tsi	+0.052	[−0.101, +0.291]	0.600	0.346
US–SB vs SB–Tsi	+0.065	[−0.115, +0.307]	0.539	0.384
US–Tsi vs SB–Tsi	+0.013	[−0.211, +0.210]	0.909	0.990



False alarms

Pairwise intergroup correlations (BCa 95% CIs)

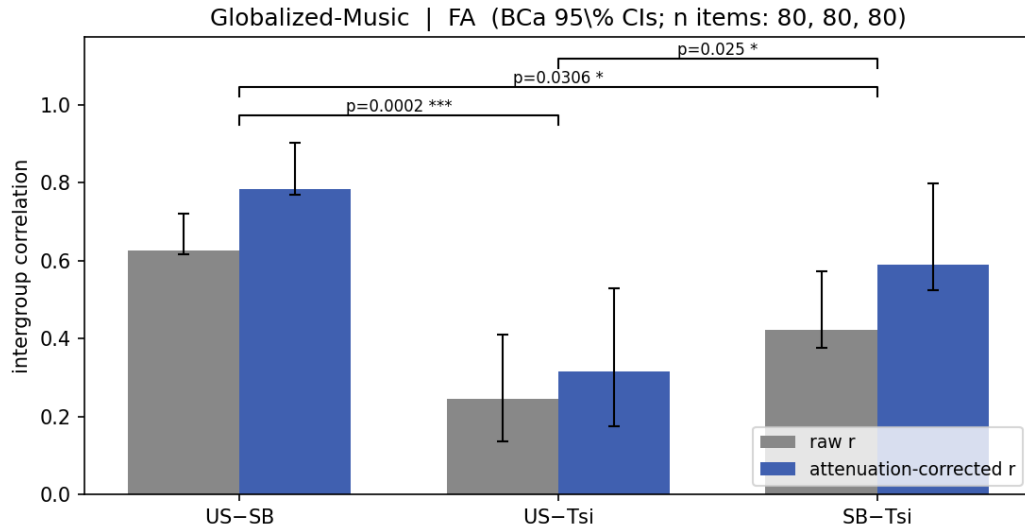
Pair	$\hat{r}$ (raw)	95% CI (BCa)	$\hat{r}^*$ (corrected)	95% CI (BCa)
US–SB	+0.628	[+0.616, +0.722]	+0.784	[+0.770, +0.902]
US–Tsi	+0.245	[+0.137, +0.410]	+0.316	[+0.176, +0.529]
SB–Tsi	+0.424	[+0.377, +0.573]	+0.591	[+0.525, +0.798]

Diagnostic — alternative CIs on corrected $\hat{r}^*$

Pair	$\hat{r}^*$	Percentile 95% CI	Fisher- z 95% CI	BCa 95% CI
US–SB	+0.784	[+0.517, +0.800]	[+0.659, +0.867]	[+0.770, +0.902]
US–Tsi	+0.316	[+0.084, +0.446]	[+0.128, +0.482]	[+0.176, +0.529]
SB–Tsi	+0.591	[+0.264, +0.659]	[+0.394, +0.735]	[+0.525, +0.798]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.383	[+0.153, +0.493]	<0.001	<0.001
US-SB vs SB-Tsi	+0.204	[+0.008, +0.384]	0.031	0.040
US-Tsi vs SB-Tsi	-0.179	[-0.281, +0.028]	0.025	0.106



Industrial-Nature

Hits

Pairwise intergroup correlations (BCa 95% CIs)

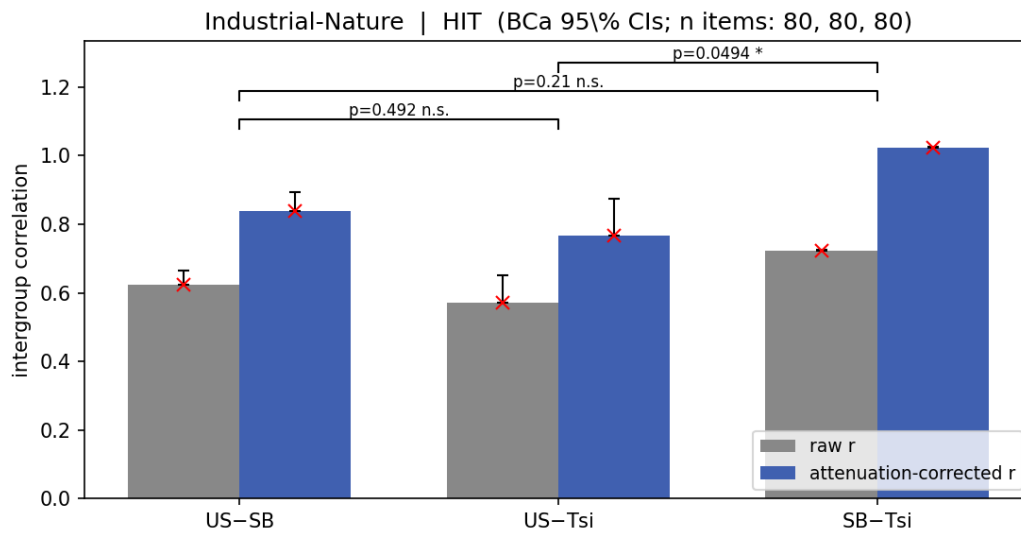
Pair	$\hat{r}$ (raw)	95% CI (BCa)	$\hat{r}^*$ (corrected)	95% CI (BCa)
US–SB	+0.623	[+0.639, +0.664]	+0.838	[+0.859, +0.893]
US–Tsi	+0.570	[+0.577, +0.650]	+0.766	[+0.776, +0.874]
SB–Tsi	+0.722	[+0.724, +0.724]	+1.022	[+1.025, +1.025]

Diagnostic — alternative CIs on corrected $\hat{r}^*$

Pair	$\hat{r}^*$	Percentile 95% CI	Fisher- z 95% CI	BCa 95% CI
US–SB	+0.838	[+0.496, +0.818]	[+0.722, +0.909]	[+0.859, +0.893]
US–Tsi	+0.766	[+0.409, +0.757]	[+0.626, +0.859]	[+0.776, +0.874]
SB–Tsi	+1.022	[+0.606, +0.940]	[+1.000, +1.000]	[+1.025, +1.025]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.053	[-0.098, +0.211]	0.492	0.483
US-SB vs SB-Tsi	-0.099	[-0.208, +0.099]	0.210	0.482
US-Tsi vs SB-Tsi	-0.152	[-0.261, +0.043]	0.049	0.162



False alarms

Pairwise intergroup correlations (BCa 95% CIs)

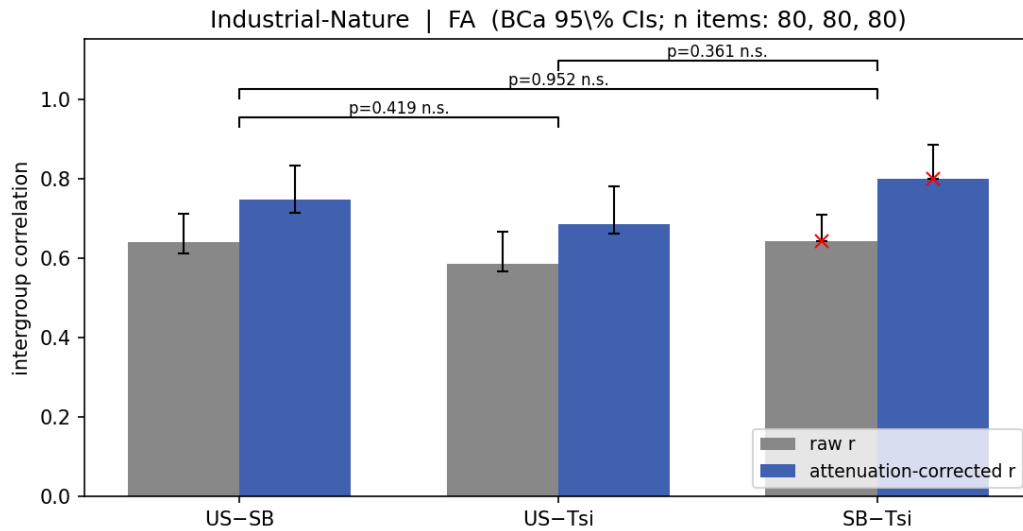
Pair	$\hat{r}$ (raw)	95% CI (BCa)	$\hat{r}^*$ (corrected)	95% CI (BCa)
US–SB	+0.640	[+0.612, +0.713]	+0.750	[+0.716, +0.835]
US–Tsi	+0.588	[+0.567, +0.668]	+0.687	[+0.663, +0.781]
SB–Tsi	+0.644	[+0.646, +0.712]	+0.802	[+0.804, +0.886]

Diagnostic — alternative CIs on corrected $\hat{r}^*$

Pair	$\hat{r}^*$	Percentile 95% CI	Fisher- z 95% CI	BCa 95% CI
US–SB	+0.750	[+0.550, +0.778]	[+0.643, +0.827]	[+0.716, +0.835]
US–Tsi	+0.687	[+0.483, +0.710]	[+0.581, +0.771]	[+0.663, +0.781]
SB–Tsi	+0.802	[+0.529, +0.797]	[+0.695, +0.874]	[+0.804, +0.886]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.053	[-0.069, +0.189]	0.419	0.344
US-SB vs SB-Tsi	-0.004	[-0.088, +0.169]	0.952	0.540
US-Tsi vs SB-Tsi	-0.056	[-0.142, +0.099]	0.361	0.699



Methods note

All correlations are Spearman. Within-group reliabilities are split-half (Spearman–Brown corrected) computed by participant split. Attenuation correction applied as $\hat{r}^* = \hat{r} / \sqrt{\hat{\rho}_A^{\text{SB}} \hat{\rho}_B^{\text{SB}}}$, clamped to $[-1, 1]$. Bootstrap CIs computed three ways from the same resampled distribution: **percentile** (the $[P_{2.5}, P_{97.5}]$ quantiles), **Fisher- z back-transformed** (normal-approx CI on the z -scale, using bootstrap z -space SD, centered on $\text{atanh}(\hat{r})$), and **BCa** (bias-corrected and accelerated with jackknife). Paired-bootstrap comparisons share the resample of any group that appears in both correlations under test; p -values are reported under two definitions, recentered-null (recommended) and straddle-zero (legacy).